

GCP Data Engineering + AI Fundamentals

Industry-Focused Training with Real-World Projects

Role: GCP Data Engineer

Course duration: 3 months

Mode: Online

Language: English

Trainer: Shaik Saidhul

GCP Cloud Basics

GCP Introduction

- The need for cloud computing in modern businesses.
- Key features and offerings of Google Cloud Platform (GCP).
- Overview of core GCP services and products.
- Benefits and advantages of using cloud infrastructure.
- Step-by-step guide to creating a free-tier account on GCP.

GCP Interfaces

- Console
 - Navigating the GCP Console
 - Configuring the GCP Console for Efficiency
 - Using the GCP Console for Service Management
- Shell
 - Introduction to GCP Shell
 - Command-line Interface (CLI) Basics
 - GCP Shell Commands for Service Deployment and Management
- SDK
 - Overview of GCP Software Development Kits (SDKs)
 - Installing and Configuring SDKs
 - Writing and Executing GCP SDK Commands

GCP Locations

- Regions
 - Understanding GCP Regions
 - Selecting Regions for Service Deployment
 - Impact of Region on Service Performance
- Zones
 - Exploring GCP Zones
 - Distributing Resources Across Zones
 - High Availability and Disaster Recovery Considerations
- Importance
 - Significance of Choosing the Right Location

- Global vs. Regional Resources
- Factors Influencing Location Decisions

GCP IAM & Admin

- Identities
 - Introduction to Identity and Access Management (IAM)
 - Users, Groups, and Service Accounts
 - Best Practices for Identity Management
- Roles
 - GCP IAM Roles Overview
 - Defining Custom Roles
 - Role-Based Access Control (RBAC) Implementation
- Policy
 - Resource-based Policies
 - Understanding and Implementing Organization Policies
 - Auditing and Monitoring Policies
- Resource Hierarchy
 - GCP Resource Hierarchy Structure
 - Managing Resources in a Hierarchy
 - Organizational Structure Best Practices

Linux Basics on Cloud Shell

- Getting started with Linux
- Linux Installation
- Basic Linux Commands
 - Cloud shell tips
 - File and Directory Operations (ls, cd, pwd, mkdir, rmdir, cp, mv, touch, rm, nano)
 - File Content Manipulation (cat, less, head, tail, grep)
 - Text Processing (awk, sed, cut, sort, uniq)
 - User and Permission related (whoami, id, su, sudo, chmod, chown)

Python for Data Engineer

- Getting Started with Python
- Introduction to Python and basic functions
- Data Types
 - Strings
 - Operators
 - Numbers (Int, Float)
 - Booleans
- Data Structures
 - Lists
 - Tuples
 - Dictionaries
 - Sets
- Python Programming Constructs
 - if, elif, else statements
 - for loops, while loops
- File I/O operations

- What is FILE IO operations
- Read, write, append, rename, remove, copy, move
- Exception Handling and Logging
 - what is exception handling, why it is needed
 - try and except blocks
 - else, finally, raise blocks
 - Why Logging?
 - Logging Levels
 - Writing Logs to File
 - Logging Best Practices
- Data and Time operations
 - Working date and time
 - Formatting
 - Parsing
 - Time delta
- Modular Programming in Python
 - Functions & Lambda Functions
 - Classes
- REST API basics
 - What is API
 - HTTP methods (GET, POST, PUT, DELETE)
 - Status Codes, Request Headers, JSON Responses
 - Using the requests Library, Reading API Responses
- Modules and Packages
 - What is module and packages
 - Importing
 - Built-in modules
 - Install packages

GCP Data Engineering Tools

Google Cloud Storage

- Overview of Cloud Storage as a scalable and durable object storage service.
- Understanding buckets and objects in Cloud Storage.
- Use cases for Cloud Storage, such as data backup, multimedia storage, and website content
- Labs: using console & CLI to do below
 - Creating and managing Cloud Storage buckets.
 - Uploading and downloading objects to and from Cloud Storage.
 - Setting access controls and permissions for buckets and objects.
 - Data Transfer and Lifecycle Management
 - Versioning and Object Versioning
- Integration with Other GCP Services
- Monitoring and logging for Cloud Storage operations.

Cloud SQL

- Introduction to Cloud SQL
- Creating and Managing Cloud SQL Instances

- Configuring database settings, users, and access controls.
- Connecting to Cloud SQL instances using Cloud SQL studio, Shell, Workbenches
- Importing and exporting data in Cloud SQL.
- Backups and High Availability
- Integration with Other GCP Services
- Managing database user roles and permissions.
- Introduction to DMS
- End to End Database migration Project
 - Manual: Export and Import method
 - Automation: Cloud SQL DMS method

BigQuery (SQL development)

- Introduction to BigQuery
- BigQuery Architecture
- Use cases for BigQuery in business intelligence and analytics.
- Various method of creating table in BigQuery
- BigQuery Data Sources and File Formats
- Native table and External Tables
- Working with complex data types
 - Working json data, nested, repeated and array data
- Data Integration and Export
 - Loading data into BigQuery from Cloud Storage, Cloud SQL, and other sources.
 - Exporting data from BigQuery to various formats.
 - Real-time data streaming into BigQuery.
- Configuring access controls and permissions in BigQuery.
- BigQuery Views:
 - Views
 - Materialized Views
 - Authorized Views
- Optimization techniques in BigQuery
- BigQuery Slots – on demand, flat-rate, flex-slots
- *Case Study-1: implement a real-world analytics data platform for Spotify*
- *Case Study-2: Enterprise Social Media analytics platform*

DataProc (Pyspark Development)

- Introduction to Hadoop and Apache Spark
- Understanding the difference between Spark and MapReduce
- What is Spark and Pyspark.
- Understanding Spark framework and its functionalities
- Overview of DataProc as a fully managed Apache Spark and Hadoop service.
- Cluster Creation and Configuration
 - Creating and managing DataProc clusters.
 - Configuring cluster properties for performance and scalability.
 - Preemptible instances and cost optimization.
- Learning Pyspark:
 - How to read from multiple data sources – csv, text, json, parquet, database table, BigQuery tables
 - How to perform multiple transformations

- How to write to multiple targets - csv, text, json, parquet, database table, BigQuery tables
- Running Jobs on DataProc
 - Submitting and monitoring Spark and Hadoop jobs on DataProc.
 - Use of initialization actions and custom scripts.
 - Job debugging and troubleshooting.
- *Case study-1: Data Cleaning of Employee Travel Records*
- *Case study-2: Processing real-time patient health data*
- *Case study-3: Creating a pyspark job to support ML model creations*

DataFlow (Apache Beam development)

- Introduction to DataFlow
- Use cases for DataFlow in real-time analytics and ETL.
- Understanding the difference between Apache Spark and Apache Beam
- How Dataflow is different from Dataproc
- Learning Apache Beam
 - How to read from multiple data sources – csv, text, json, parquet, database table, BigQuery tables
 - How to perform multiple transformations
 - How to write to multiple targets - csv, text, json, parquet, database table, BigQuery tables
- *Case study-1: Template method of creating pipelines*
- *Case study-2: E-commerce Transaction Processing*
- *Case study-3: End to End Streaming Pipeline using Apache beam with Dataflow, Python app, PubSub, BigQuery, GCS*

Cloud Pub/Sub (Streaming)

- Introduction to Pub/Sub
- Understanding the role of Pub/Sub in event-driven architectures.
- Key Pub/Sub concepts: topics, subscriptions, messages, and acknowledgments.
- Creating and Managing Topics and Subscriptions
 - Using the GCP Console to create Pub/Sub topics and subscriptions.
 - Configuring message retention policies and acknowledgment settings.
- Publishing and Consuming Messages
 - Writing and deploying code to publish messages to a topic.
 - Implementing subscribers to consume and process messages from subscriptions.
- *Case study-1: Streaming use-case using Dataflow*

Cloud Composer (DAG Development)

- Introduction to Composer/Airflow
- Overview of Airflow Architecture
- Use cases for Composer in managing and scheduling workflows.
- Creating and Managing Workflows
 - Creating and configuring Composer environments.
 - Defining and scheduling workflows using Apache Airflow.
 - Monitoring and managing workflow executions.
- Integration with Data Engineering Services
 - Orchestrating workflows involving BigQuery, DataFlow, and other services.

- Coordinating ETL processes with Composer.
- Integrating with external systems and APIs.
- Error Handling and Troubleshooting
 - Handling errors and retries in Composer workflows.
 - Debugging and troubleshooting failed workflow executions.
 - Logging and monitoring for Composer workflows.
- *Level-1-DAG: Orchestrating the BigQuery pipelines*
- *Level-2-DAG: Orchestrating the DataProc pipelines*
- *Level-3-DAG: Orchestrating the Dataflow pipelines*
- *Deploy DAGs: Implementing CI/CD in Composer Using Cloud Build and GitHub*

Databricks on GCP

- a. What is Lakehouse Architecture?
- b. Difference Between:
 - Data Lake
 - Data Warehouse
 - Data Lakehouse
- c. Introduction to Databricks
 - Overview of Databricks
 - Databricks Architecture
 - Setting up a Databricks Workspace
- d. Unity Catalog - Unified Data Governance
 - Introduction to Unity Catalog
 - Core Concepts: Metastore, Catalog, Schema, Tables, Volumes, Functions, Models
 - External Data Access
 - a. Storage Credentials
 - b. External Locations (S3, GCS, ADLS)
 - Lakehouse Federation - Foreign Catalog, Connection
 - Delta Sharing – Share, Recipient, Provider
- e. Databricks Clusters
 - Introduction to Clusters
 - Types of Clusters
 - Cluster Modes
- f. DBUtils Commands - File handling, Notebook Widgets, Secrets
- g. Introduction to Delta Lake
 - What is Delta Lake?
 - Creating Delta Lake Tables - Managed Tables, External Tables
 - Understanding Delta Lake Table Creation - Delta Log, Parquet Data Files
 - Advanced Delta Lake Features
 - a. Versioning
 - b. Time Travel
 - c. Compacting
 - d. Liquid clustering
 - e. Vacuum
- h. Databricks Jobs & Pipelines
 - Creating a Job
 - Scheduling Jobs
 - Setting Parameters

- Managing Dependencies
- Setting up Alerts
- i. Case study-1: Structured Streaming with Databricks
- j. Case study-2: Incremental Data Loading with Auto Loader

Data Fusion (complementary)

- Introduction to Data Fusion
 - Overview of Data Fusion as a fully managed data integration service.
 - Use cases for Data Fusion in ETL and data migration.
- Building Data Integration Pipelines
 - Creating ETL pipelines using the visual interface.
 - Configuring data sources, transformations, and sinks.
 - Using pre-built templates for common integration scenarios.
- Integration with GCP and External Services
 - Integrating Data Fusion with BigQuery, Cloud Storage, and other GCP services.
- *Case study-1: End to End pipeline using Data fusion with Wrangler, GCS, BigQuery*

Cloud Functions (complementary)

- Cloud Functions Introduction
- Setting up Cloud Functions in GCP
- Event-driven architecture and use cases
- Writing and deploying Cloud Functions
- Triggering Cloud Functions:
 - HTTP triggers
 - Pub/Sub triggers
 - Cloud Storage triggers
- Monitoring and logging Cloud Functions
- *Usecase-1: Loading the files from GCS to BigQuery as soon as it is uploaded.*

Terraform (complementary)

- Terraform Introduction
- Installing and configuring Terraform.
- Infrastructure Provisioning
- Terraform basic commands
 - Init, plan, apply, destroy
- Labs: Create Resources in Google Cloud Platform
 - GCS buckets
 - Dataproc cluster
 - BigQuery Datasets and tables
 - And more resources as needed

AI Fundamentals for GCP Data Engineers (Current Market Demand)

- Introduction to Artificial Intelligence & Generative AI
 - Understanding AI, ML, Deep Learning, Generative AI, and LLMs.
 - Real-world AI use cases in Data Engineering and Analytics.
- Machine Learning Fundamentals
 - Supervised, Unsupervised, and Reinforcement Learning.
 - Training, validation, testing, and model evaluation concepts.

- Prompt Engineering for Data Engineers
 - Designing effective prompts for Gemini, ChatGPT, and Vertex AI.
 - Using AI for SQL generation, code generation, documentation, and data analysis.
- Google Vertex AI Fundamentals
 - Overview of Vertex AI ecosystem.
 - Working with foundation models, Model Garden, and Gemini APIs.
- AI-Powered Data Pipelines
 - Integrating AI services with BigQuery, GCS, Dataflow, and Dataproc.
 - Building intelligent ETL/ELT pipelines using AI-driven transformations.

Complementary End-to-End Projects:

1. **Healthcare project on GCP of 8+ hours**
2. **Road Traffic project on Databricks of 5+ hours**

By the End of the course What Students can Expect

- ✓ **SQL Development:** Master SQL for BigQuery and Cloud SQL, including complex queries, optimization, and efficient schema design.
- ✓ **PySpark Development:** Develop and optimize scalable PySpark applications using Spark RDDs, Dataframes, and transformations.
- ✓ **Apache Beam:** Build parallel data processing pipelines and integrate Beam with Google Cloud services for scalable ETL.
- ✓ **Airflow & DAGs:** Design and manage Airflow DAGs to orchestrate end-to-end workflows and automate GCP data pipelines.
- ✓ **Databricks & Delta Lake:** Build scalable data pipelines, work with Delta Lake, Databricks SQL, notebooks, workflows, and performance optimization.
- ✓ **Data Architecture:** Design scalable, reliable, secure, and cost-effective end-to-end data engineering architectures on Google Cloud.
- ✓ **Certification + ML/AI Readiness:** Prepare for Google Cloud PDE & ACE certifications while learning how BigQuery, Vertex AI, and Gemini integrate with modern AI-enabled data platforms.

Course Highlights:

- Hands-on Labs & Real-world Use Cases
- Live Q&A Sessions
- Course Completion Certificate
- Access to Study Materials & Code Repository
- Industry-Standard Best Practices

The course will empower students with practical skills in SQL, PySpark, Apache Beam, DAG creations, and architecture planning, ensuring they are well-prepared to tackle real-world data engineering challenges and successfully obtain GCP certifications.

Thank You.